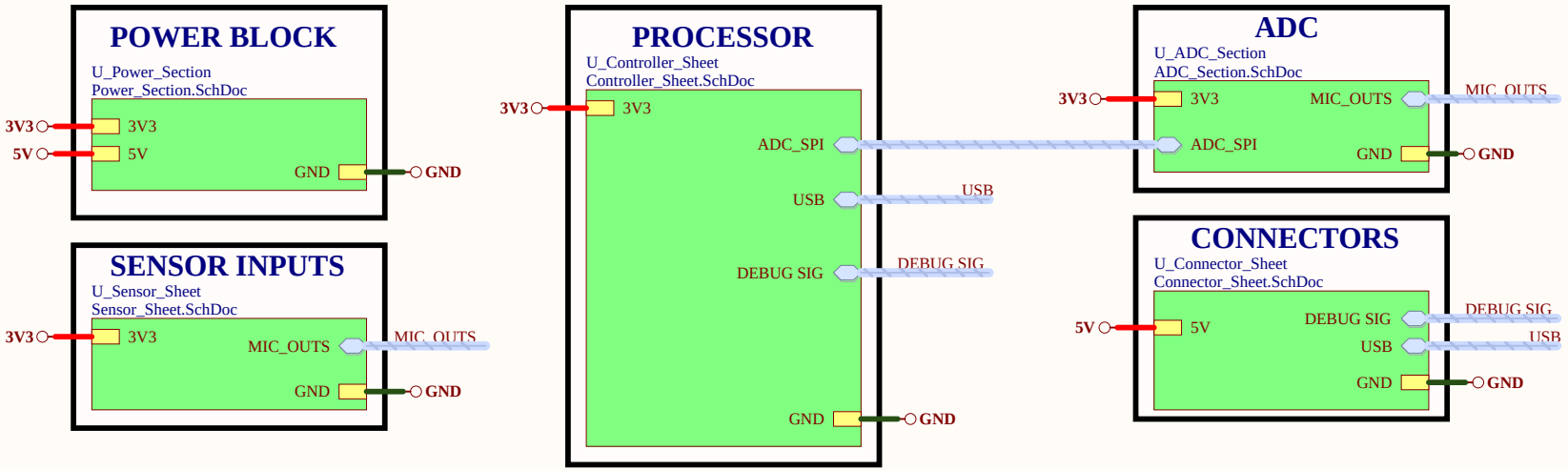
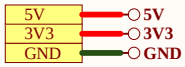
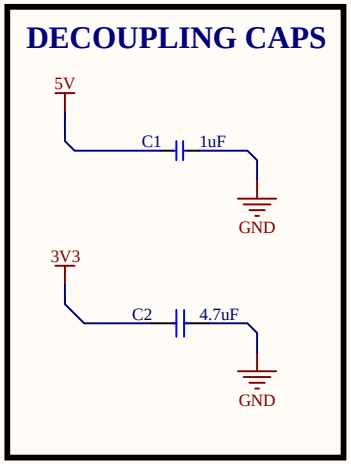
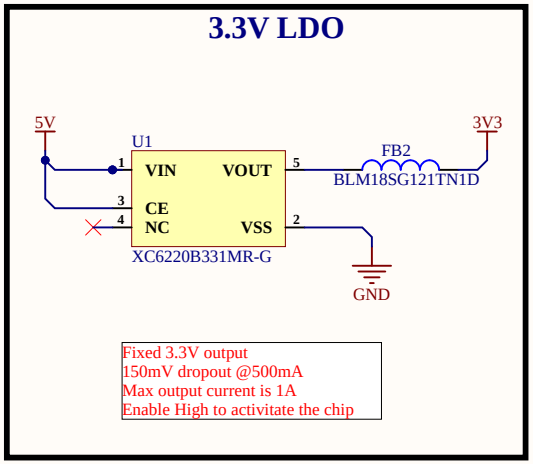


BLOCK_DIAGRAM

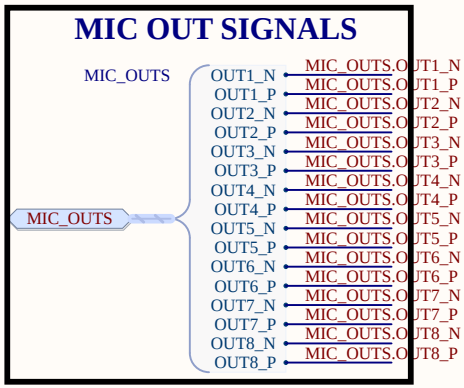


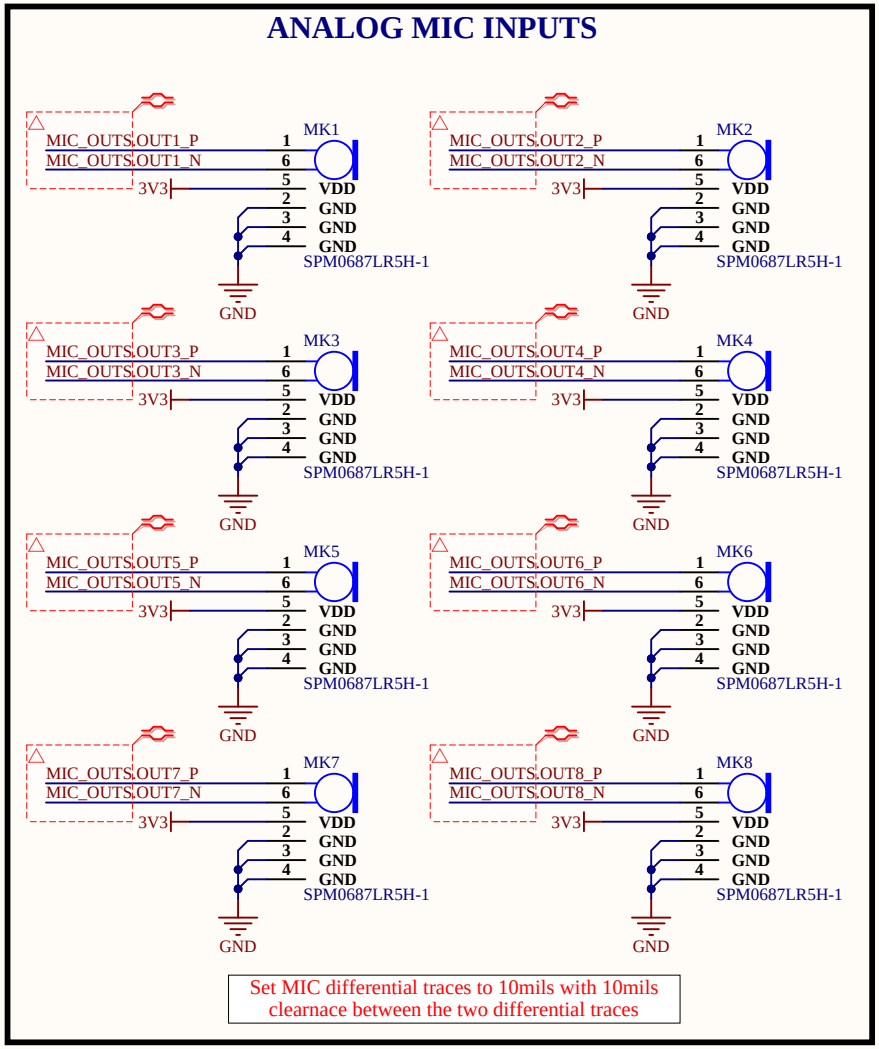


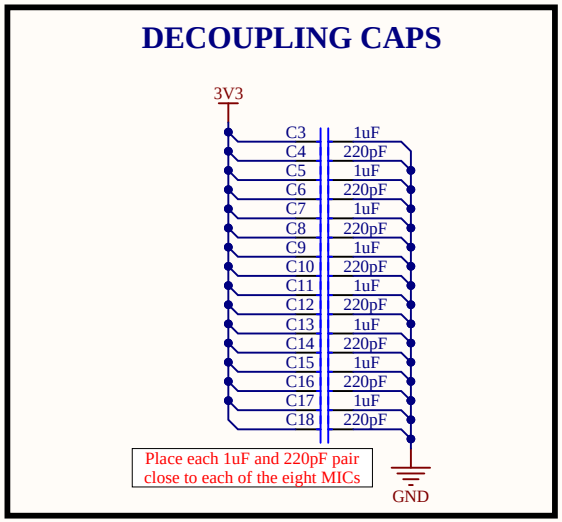


POWER_SECTION

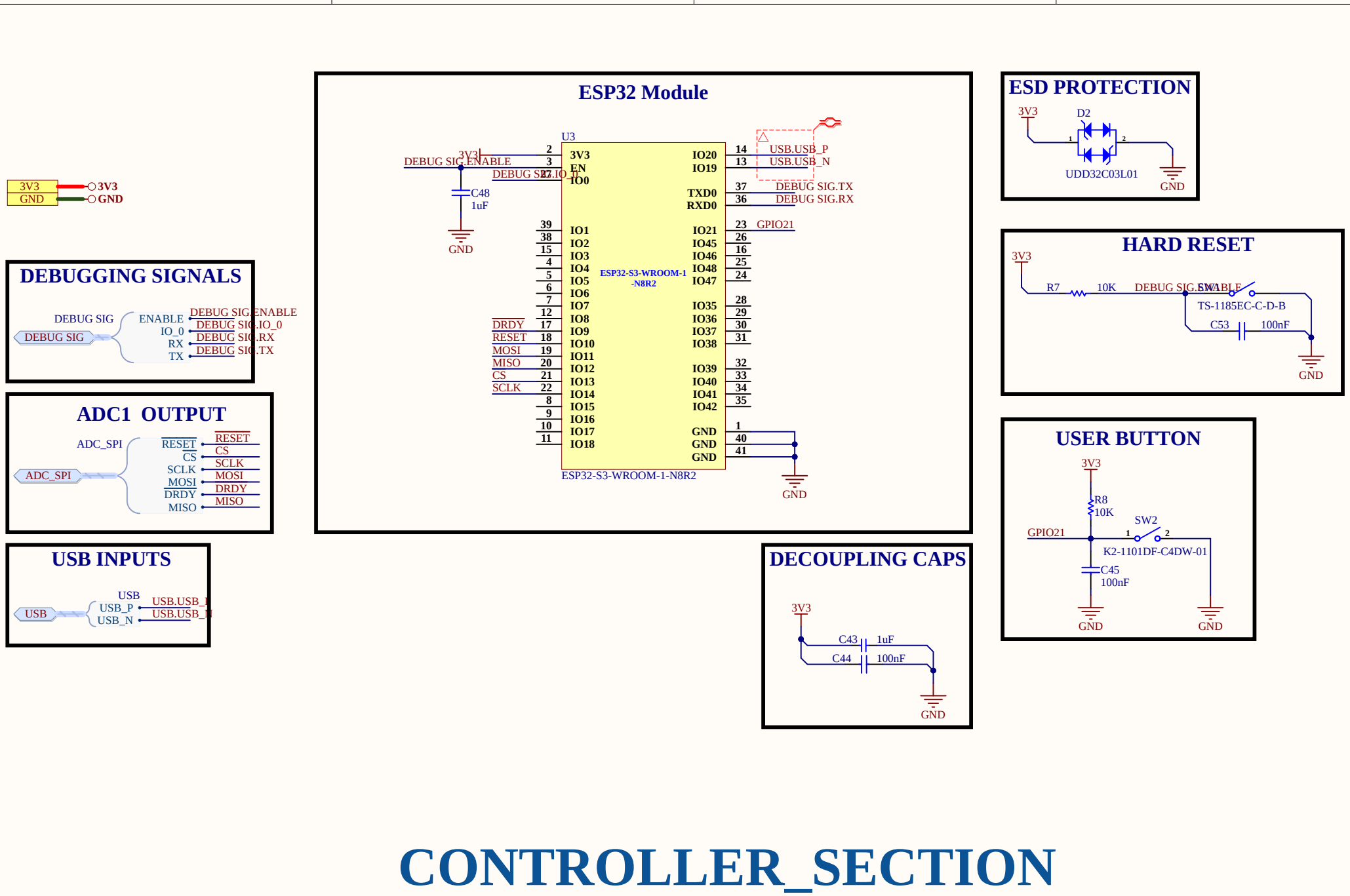




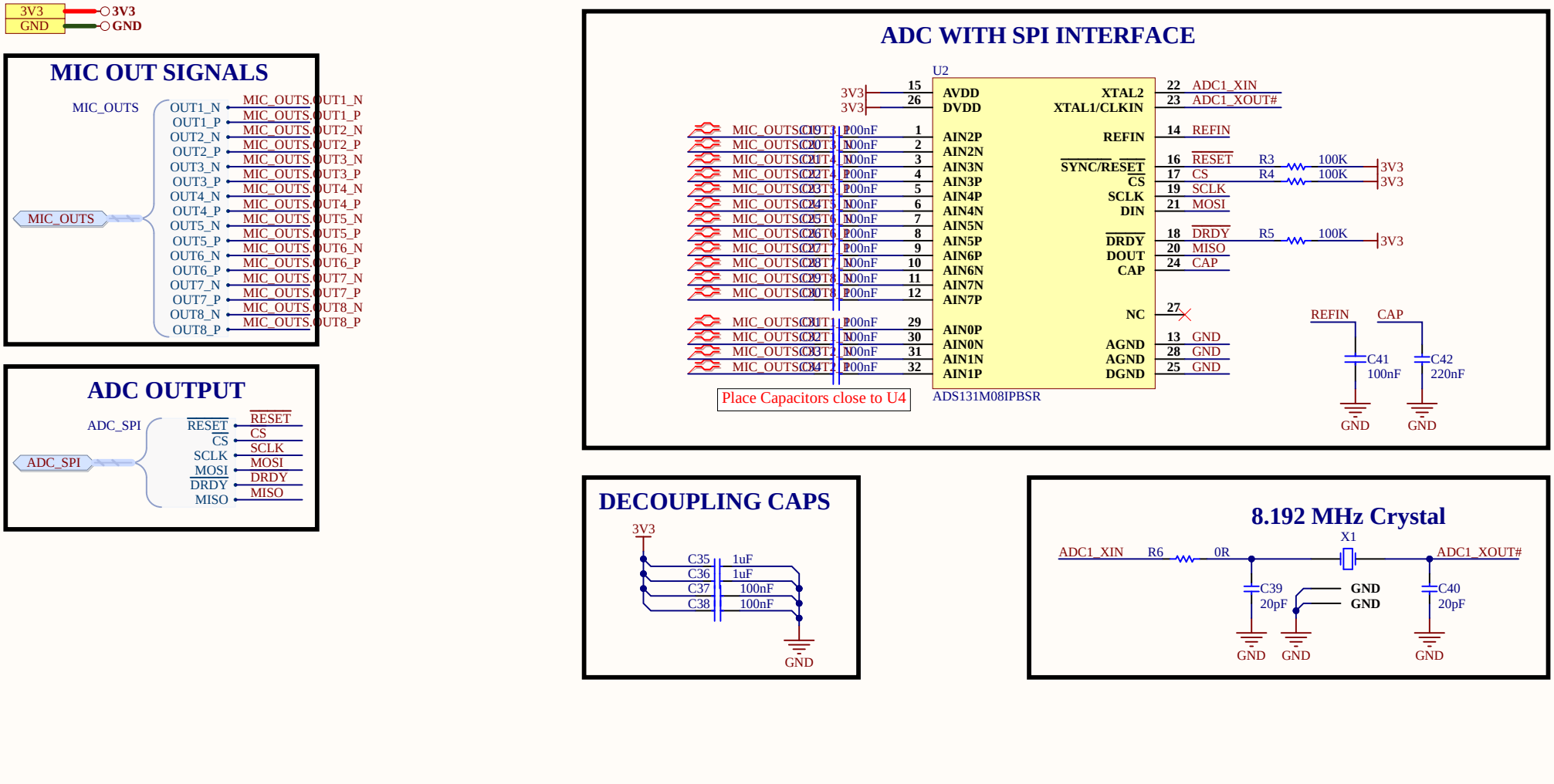




SENSOR_SECTION



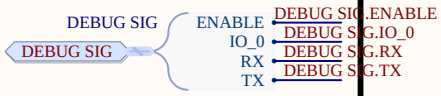
CONTROLLER_SECTION


3V3
GND
MIC OUT SIGNALS
MIC_OUTS
OUT1_N
OUT1_P
OUT2_N
OUT2_P
OUT3_N
OUT3_P
OUT4_N
OUT4_P
OUT5_N
OUT5_P
OUT6_N
OUT6_P
OUT7_N
OUT7_P
OUT8_N
OUT8_P
MIC_OUTS.OUT1_N
MIC_OUTS.OUT1_P
MIC_OUTS.OUT2_N
MIC_OUTS.OUT2_P
MIC_OUTS.OUT3_N
MIC_OUTS.OUT3_P
MIC_OUTS.OUT4_N
MIC_OUTS.OUT4_P
MIC_OUTS.OUT5_N
MIC_OUTS.OUT5_P
MIC_OUTS.OUT6_N
MIC_OUTS.OUT6_P
MIC_OUTS.OUT7_N
MIC_OUTS.OUT7_P
MIC_OUTS.OUT8_N
MIC_OUTS.OUT8_P
ADC OUTPUT
ADC_SPI
RESET
CS
SCLK
MOSI
DRDY
MISO
ADC WITH SPI INTERFACE
U2
AVDD
DVDD
AIN2P
AIN2N
AIN3N
AIN3P
AIN4P
AIN4N
AIN5N
AIN5P
AIN6P
AIN6N
AIN7N
AIN7P
AIN0P
AIN0N
AIN1N
AIN1P
XTAL2
XTAL1/CLKIN
REFIN
SYNC/RESET
CS
SCLK
DIN
DRDY
DOUT
CAP
NC
AGND
AGND
DGND
ADC1_XIN
ADC1_XOUT#
REFIN
RESET
CS
SCLK
MOSI
DRDY
MISO
CAP
GND
GND
GND
GND
100K
100K
100K
3V3
3V3
3V3
100nF
ADS131M08IPBSR
Place Capacitors close to U4
C41
100nF
C42
220nF
DECOUPLING CAPS
C35
1uF
C36
1uF
C37
100nF
C38
100nF
8.192 MHz Crystal
X1
R6
0R
C39
20pF
C40
20pF

ADC_SECTION



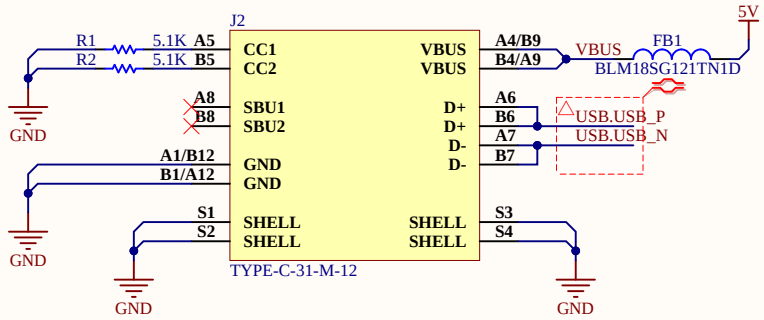
DEBUGGING SIGNALS



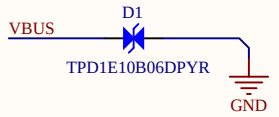
USB INPUTS



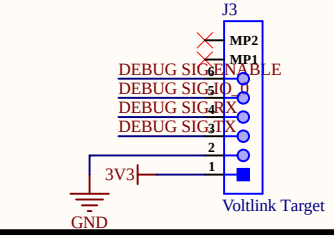
5V INPUT



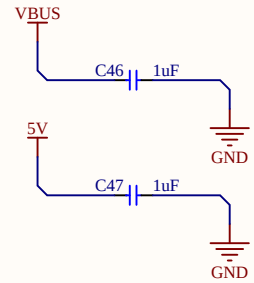
ESD PROTECTION



DEBUG CONNECTOR



DECOUPLING CAPS



CONNECTOR_SECTION